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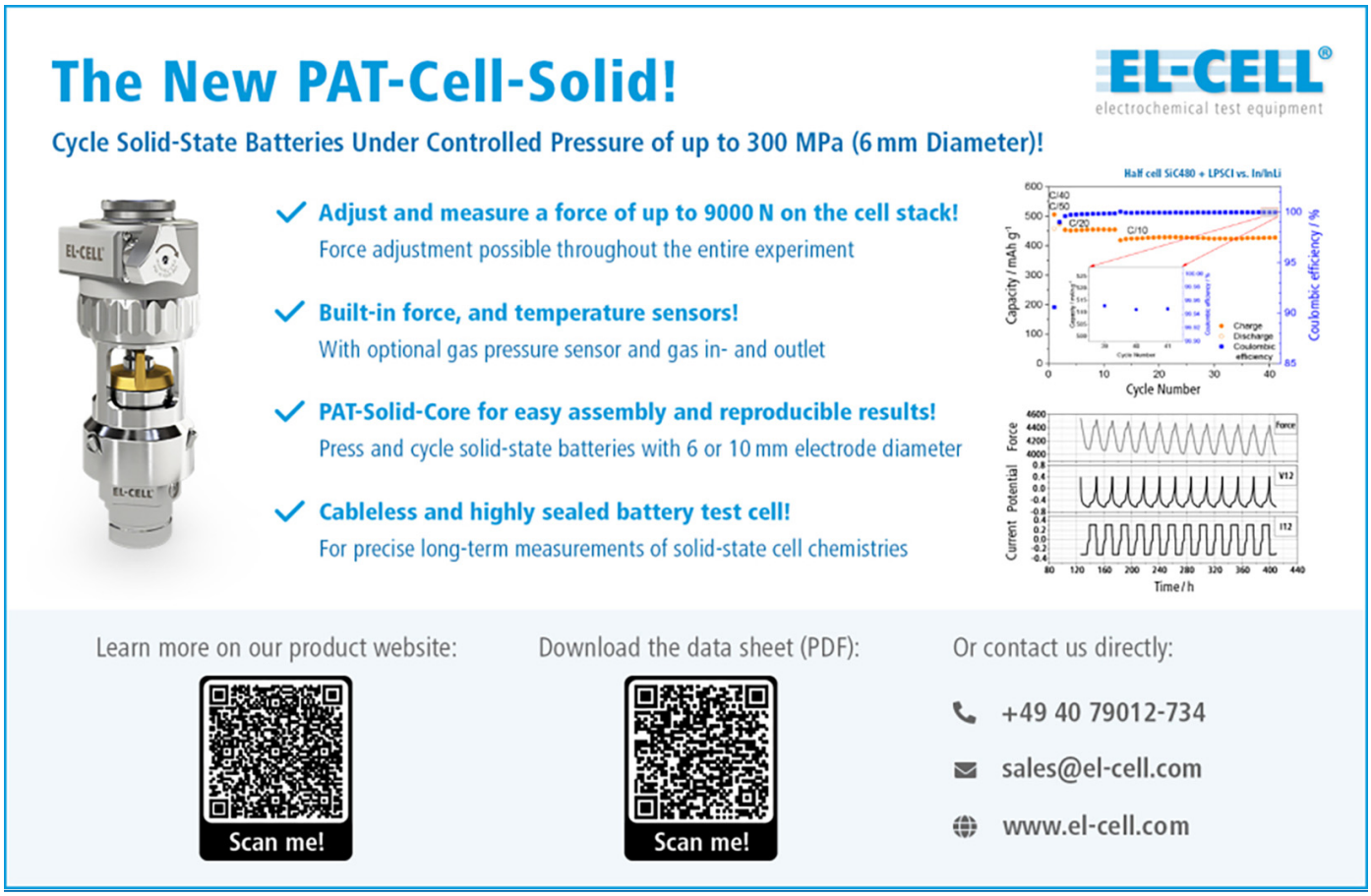
# Facile and Very Sensitive Electrochemical Method for Evaluating the Release Kinetics of Caffeine from Bioactive Polymeric Scaffolds

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# Facile and Very Sensitive Electrochemical Method for Evaluating the Release Kinetics of Caffeine from Bioactive Polymeric Scaffolds

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In the presented research facile, sensitive and accurate electrochemical method i.e. differential pulse voltammetry is described for evaluating the release kinetics of caffeine (CAF) from 3D bioactive polymeric scaffolds with application in bone tissue engineering. For the preparation of composite filaments (polylactide - caffeine) in a mass ratio 9:1 and 3D scaffolds Filastruder v1.0 extruder and RepRapPro Tricolor Mendel 3D printer were applied. All produced 3D structures fully reproduce virtual models. The amounts of caffeine released from filaments and scaffolds were analyzed for 96 h in deionized water, 0.1 mol · $L^{-1}$ phosphate buffer saline (PBS, pH 7.4) and 0.1 mol · $L^{-1}$ HCl. UV-Vis spectrophotometry was used as a reference method according to the recommendation of European pharmacopoeia (9th ed., 2016). For both methods selectivity, sensitivity, precision and accuracy were specified. Differential pulse voltammetry technique has proved to be competitive and to have high sensitivity, greater range of linearity, short analysis time, easy operation and low cost meeting the requirements of green analytical chemistry contrary to UV-Vis spectrophotometry.
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Application of the polymer scaffolds is one of the most effective medical procedure for treatment of mechanical damages of various tissues especially bone and cartilage. Three-dimensional form supports cells migration, growth and proliferation. The structure described by shape, dimensions, infill and the material properties dependent on the main component and the additives should be adequate to the type of damage and to the type of tissue.[1,2] Regeneration of the tissue can be supported by bioactive molecules distributed in polymer matrix. The scaffold as a local delivery system facilitates the process of healing by releasing substances at an appropriate rates and concentrations. This procedure is widely used in vascular[3,4] or bone and cartilage[5,6] tissue engineering. Controlled quantitative incorporation of the bioactive molecules into filaments, providing of their homogeneous distribution in prepared scaffolds and, in particular, analysis of their release kinetics and effectiveness are the key problems facing modern tissue engineering.

Natural polymers including: collagen, fibrin, chitosan, alginates, hyaluronates etc. and synthetic polymers like: polylactide (PLA), polyglicolide, polycaprolactone etc. are the basic materials used in tissue engineering.[7–10] Desired properties are usually achieved by preparation of the two- or multi-component physical mixture of homopolymers or copolymers slightly modified by addition of the other compounds.[11,12] New materials could be processed to obtain three-dimensional scaffolds using a wide range of methods where the most popular are solvent casting and particulate leaching,[13,14] freeze drying,[15,16] phase separation,[17,18] gas foaming[19,20] and rapid prototyping.[21,22]

One of the most interesting and promising method of polymer processing is 3D printing by fused deposition modelling (FDM).[23,24] Thermoplastic material in form of a filament is melted and casted layer by layer to produce three dimensional object with structure predefined by the computer model.[25] The most important advantages compare to the traditional methods include easy and direct control of microstructure including porosity or pore distribution, lower costs and shorten time of the process. The resolution of produced object and the range of available materials could be some limitation depending on the requirements.

Polymer scaffold loaded with bioactive additive could be produced by FDM printing in 3D after the preparation of filament with an adequate composition. The substance should be homogeneously distributed and stable at the temperature of processing. A simple example of this type of compound as a model for 3D printing materials is caffeine.[26] Caffeine (3,7-dihydro-1,3,7-trimethyl-1H-purine-2,6-dione) is a purine alkaloid organic compound present in coffee beans, yerba mate leaves, tea, guarana seeds, cola nuts and cacao beans.[27] It affects the central nervous system, accelerates the metabolism, increases blood pressure, reduces sleepiness and improves mood and concentration.[28] Excessive concentrations of caffeine, however, induce hyperactivity, trembling, heart disease, nausea, vomiting, depression and mutation effects such as inhibition of DNA.[29] It helps in the treatment of asthma, nasal congestion, headaches, migraines, it reveals the tension and increases the production of dopamine. Addition of this compound enhances the performance of many medicaments. Therefore, it is widely used in nonsteroidal pharmaceuticals. Caffeine as a neurological stimulant has a significant influence on tissue activity,[30,31] hence it was used in this research as a bioactive compound for the preparation of polylactide scaffolds.

Various analytical methods have been used for the detection and quantification of CAF including gas chromatography,[30] high performance liquid chromatography,[32] spectrophotometry,[33,34] voltammetry[28,29,35–37] and amperometry.[38] Chromatographic and spectroscopic methods involve sophisticated instrumentation, time consuming and complicated sample pre-treatments and complex extraction which make them unsuitable for on-site monitoring. Electrochemical methods, especially differential pulse voltammetry (DPV), have been widely recognized as a powerful tool due to its high sensitivity, short analysis time, easy operation, low cost, continuous real-time detection and good portability. Working electrodes employed for the determination of CAF usually include modified and non-modified carbon electrodes,[27,28,36,37,39] carbon paste electrodes,[40] carbon-ceramic electrodes,[41] carbon-fiber ultramicroelectrodes[29] and boron-doped diamond electrodes.[35]

The main aim of this paper was to develop and optimize a fast and simple voltammetric procedure employing glassy carbon electrode (GCE) and DPV technique for evaluation of the release kinetics and effectiveness of CAF from 3D bioactive polymeric scaffolds in extracts samples that were not pretreated. Analyses of CAF release from the scaffolds were performed in high-purity deionized (DI) water, 0.1 mol · $L^{-1}$ phosphate-buffer saline (PBS, pH 7.4) and 0.1 mol · $L^{-1}$ HCl. UV-Vis spectrophotometry was used as a reference method according to recommendation of the European pharmacopoeia (9th ed., Strasbourg, Council of Europe, 2016).[42] As per our knowledge this is first report where voltammetry technique is used in polymer tissue engineering.

## Experimental

***Reagents and materials.—***All chemicals were of analytical grade and used without further purification. Caffeine (≥99.0%) was purchased from Sigma Aldrich, USA. The stock standard solution of CAF (0.01 mol · $L^{-1}$) and working solutions were prepared daily by dissolving a reagent or a standard solution in DI water or dichloromethane.

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Concentrated sulfuric acid(VI) (98% w/v), hydrochloric acid (37% w/v) and dichloromethane were obtained from Merck. Phosphate buffered saline (PBS) was prepared by dissolving: 5.000 g of sodium chloride, 0.125 g of potassium chloride, 0.900 g of disodium hydrogen phosphate and 0.150 g of potassium dihydrogen phosphate in 500 mL of DI water. Commercial polylactide granulate for the preparation of filament was purchased from Barrus, Holland. High-purity deionized water ($>18$ M$\Omega \cdot$cm) obtained from the Milli-Q water purification system (Millipore, UK) was used for the preparation of all aqueous solutions. Glassware was first immersed in 6 M hot nitric acid, next rinsed repeatedly with DI water.

***Instrumentation and apparatus.—***All voltammetric measurements were performed using M161 multipurpose electrochemical analyser equipped with M164 electrode stand (both mtm-anko, Kraków, Poland). Data acquisition and experimental control were processed with EALab 2.1 software.[43] A quartz cell of 5 mL volume was used with a conventional three-electrode system with 3 mm diameter glassy carbon electrode (GCE, BASi, USA) as working electrode, Ag/AgCl/3 mol $\cdot$ L$^{-1}$ KCl electrode with replaceable outer junction (2.5 mol $\cdot$ L$^{-1}$ $KNO_3$) as a reference and 1 mm diameter Pt wire as an auxiliary electrode. Before each series of experiments, the GCE was polished with aluminum slurries of successively decreasing final grades down to 0.05 μm (Buehler, USA) on polishing cloths (Buehler, USA) and subsequently rinsed with water, methanol and again with water. Next, the electrode was electrochemically activated in supporting electrolyte solution (0.05 mol $\cdot$ L$^{-1}$ $H_2SO_4$) using scanning electrode potential ($\nu$ = dE/dt = 0.05 V $\cdot$ s$^{-1}$) in the range between $-0.2$ and 1.8 V vs. Ag/AgCl (in terms of hydrogen and oxygen evolution) until a stable cyclic voltammogram (CV) was obtained. Satisfactory precision of quantitative CAF determination was achieved by application of additional, short polarization of the GCE electrode in the analyzed solution at potential $-0.2$ V for 3 s, directly before each measurement (DP voltammogram registration).[44] UV-Vis spectra were taken using Lambda XLS UV-Vis spectrometer (PerkinElmer, USA) with 2 nm slit width and 1 cm quartz cuvette. PLA/CAF scaffolds were prepared with RepRapPro Tricolour Mendel 3D printer (RepRapPro, United Kingdom). Images of the scaffolds were taken using Keyence VHX-900F digital microscope (Keyence, Japan). CPI-505 laboratory pH meter (ELMETRON, Poland) was used to verify the pH of the prepared phosphate buffer saline (PBS). All measurements were carried out at the room temperature.

***Preparation of PLA/CAF filament.—***PLA/CAF (mass ratio 9:1) filament was prepared with Filastruder v1.0 (Filastruder, USA). This simple model of the single-screw extruder is designed to a small-scale fabrication of filaments with a diameter depending on the used nozzle. The rotational speed of the screw is fixed and working temperature can be changed up to 260°C according to the producer. Diameter and density of the prepared filament was measured using caliper and pycnometer respectively. The content of CAF was determined spectrophotometrically at 278 nm using Lambda XLS UV-Vis spectrometer (PerkinElmer, USA) after dissolution of the 100 mg filament in 50 mL of dichloromethane.

***Preparation of PLA/CAF scaffolds.—***PLA scaffolds loaded with CAF were made with RepRapPro Tricolour Mendel (RepRapPro, United Kingdom) 3D printer by fused deposition modeling. The device works using Cartesian coordinate system. Printed object is formed on a heating bed covered with Kapton tape. The filament is delivered by an extruder to a hot end with a 0.5 mm nozzle. This commercially available printer was designed by open-source project RepRap. Blender, Sli3er and Pronterface were used as a free GNU GPL set of software to design the structure of the scaffold and to optimize and conduct the process of printing. Accuracy and repeatability of the method were evaluated by measurement of mass and dimension of the printed scaffolds. They were also analyzed with Keyence VHX-900F digital microscope (Keyence, Japan).

***Solid-liquid extraction of CAF.—***In vitro release assay was assigned to one selected series of nine scaffold samples. The environments applied for storage and simultaneously mimicking of the human body conditions were DI water, 0.1 mol $\cdot$ L$^{-1}$ phosphate buffer saline (pH 7.4) and 0.1 mol $\cdot$ L$^{-1}$ HCl. Scaffolds were put in closed polypropylene containers and poured with 10 mL of the appropriate diluent (three samples for each solution). Next, they were placed into Memmert INE 400 laboratory oven (Memmert, Germany) at human body temperature of $37.0 \pm 0.1$°C. After 1h solutions were transferred quantitatively to the flasks and CAF was determined immediately and simultaneously by DPV and UV-Vis spectrometry. Scaffolds were dried on filter paper, put again in empty containers and poured with 10 mL of the same fresh diluent. This procedure was repeated after 1, 2, 4, 24, 48, 72 and 96 hours.

***Standard procedure of CAF voltammetric determination.—***Quantitative measurements of CAF were performed using DPV technique and standard addition procedure. An appropriate volume of standard CAF or sample solution and 0.25 mL of 1 mol $\cdot$ L$^{-1}$ sulfuric acid(VI) were added to the voltammetric quartz cell and filled with DI water up to 5 mL. DPV procedure was performed in continuous procedure with the following steps: (i) short electrochemical cleaning of the GCE (application of the activation potential $E_{act} = -0.20$ V vs. Ag/AgCl/3 mol $\cdot$ L$^{-1}$ KCl for the activation time $t_{act} = 3$ s with stirring); (ii) equilibration time ($t_{eq} = 3$ s); (iii) recording of the DP voltammogram (in the anodic direction from 1.00 to 1.60 V with a step potential $E_s = 3$ mV, pulse amplitude dE = 50 mV and pulse period (width) $t_{imp} = (t_w + t_s) = 20$ ms; $t_w = t_s$ (waiting time = current sampling time)).

At least three independent assays were carried out for all experiments and a minimum of three repetitive scans were made per measurement.

***Spectrophotometric (UV) measurements.—***UV-Vis spectrophotometry was applied as a reference method for the quantitative determination of CAF to compare with the proposed DPV method. Spectrophotometric measurements were performed according to the procedure drawn from European pharmacopoeia (2016).[42,45] Absorbance of the solutions was measured by UV-Vis spectrophotometer at room temperature. Molar decadic absorption coefficients of CAF in dichloromethane and DI water were obtained.

## Results and Discussion

***Preparation of the PLA/CAF filament.—***PLA and CAF were mixed in a mass ratio 9:1 (optimal for planned research) and thermally processed three times at the temperature of 185°C. During the first step both dry substances were preliminary mixed and melted. Two more steps include milling and repeated melting of the material to provide a high homogeneity of PLA/CAF composite filament. This procedure was necessary to get an appropriate quality of the printed scaffolds. 10% w/w content of CAF was determined as an optimal by changing its content between 0 and 20% w/w. Mechanical properties of the filament at the higher CAF concentrations were not sufficient to prevent breaking so manipulation with the material during printing was highly difficult. Density and diameter of the prepared filament were measured as $1.271 \pm 0.009$ g $\cdot$ cm$^{-3}$ and $1.64 \pm 0.07$ mm respectively. Standard commercially available filaments have diameter tolerance between $\pm 0.05$ and $\pm 0.10$ mm so the quality of prepared one is proper. The real content of CAF in the prepared PLA/CAF composite i.e. $8.3 \pm 0.9$% w/w was determined based on the previous calibration curve. For this, 100 mg of the filament was dissolved in 50 mL of dichloromethane and analyzed by UV-Vis spectrophotometry. This procedure were repeated three times for the samples taken from the different parts of the filament. The lower than expected 10% w/w CAF content probably could be a result of a small loss of CAF powder during the preliminary mixing.

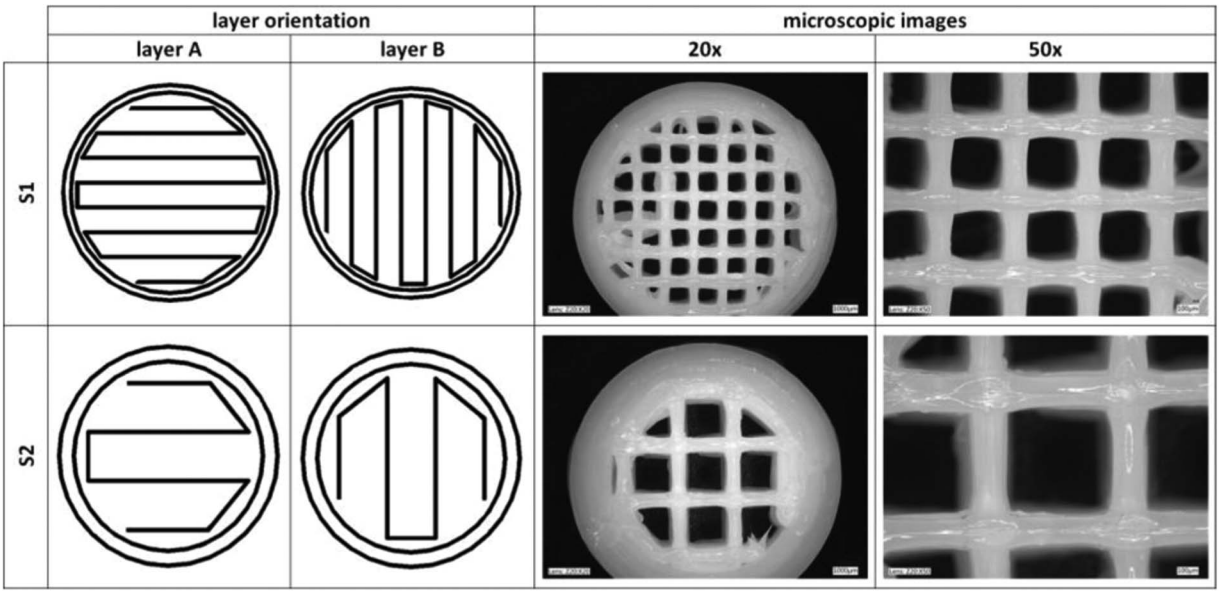



**Figure 1.** Exemplary structures of the prepared scaffolds.

***Design model of the scaffolds.—***Three dimensional virtual models of the scaffolds were designed using Blender software. An appropriate shape was chosen based on the literature review and based on the typical structure and dimensions of implants used in medicine. The model was an external shape of cylinder with 15.0 mm of the diameter and 5.4 mm of the height. The inner structure was built of solid and porous parts. The first one consist of two perimeters and the second one was constructed of parallel lines with 90° orientation between layers (ABAB arrangement, Fig. 1). The height of the layer was 0.3 mm and the thickness of the perimeter and the line was equal to the extrusion width. Four model (S1-S4, Table I.) were designed as a combination of two parameters - porosity (50 or 70% equivalent to 50 or 30% infill respectively) and extrusion width (0.5 or 1.0 mm).

***Optimization of the scaffolds printing process.—***Quality of the prepared scaffolds has been affected by a vast range of the printing process parameters. They should be precisely optimized using Sli3er software. The basic and the most important include nozzle and bed temperatures and printing movements speed. Temperature of the nozzle depends on the type of processed thermoplastic polymer and should provide a stable flow of the melted material. A risk of filament decomposition at higher temperatures should be also taken into consideration. For PLA/CAF filament the temperature of 200°C was applied. Heating of the bed supports adhesion of the printed scaffold but deformation of the lowest layers could be caused by increased temperature. After optimization of the other parameters the bed temperature of 20°C was sufficient for the designed models. Total time of the scaffold preparation depends on the speed of printing movements which was 10 $mm \cdot s^{-1}$ for used material and models. At the higher values melted material is not casted uniformly and at the lower values the printed parts are damaged by long proximity of the hot nozzle.

**Table I. Parameters of the designed scaffolds models.**

| Model | Diameter [mm] | Height [mm] | Layer height [mm] | Porosity [%] | Extrusion width [mm] |
|---|---|---|---|---|---|
| S1 | 15.0 | 5.4 | 0.3 | 70 | 0.5 |
| S2 | 15.0 | 5.4 | 0.3 | 70 | 1.0 |
| S3 | 15.0 | 5.4 | 0.3 | 50 | 0.5 |
| S4 | 15.0 | 5.4 | 0.3 | 50 | 1.0 |

***Printing of the scaffolds.—***The process of printing was controlled using Pronterface software. Two scaffolds of the one type were printed simultaneously. Total time of the process varied from about 10 to 17 minutes depending on the model.

***Quality check of the prepared scaffolds.—***Quality of the printed scaffolds was evaluated using the series of ten samples for each model. Real porosity was calculated based on dimensions and mass measurements and based on material density. All results are summarized in Table II.

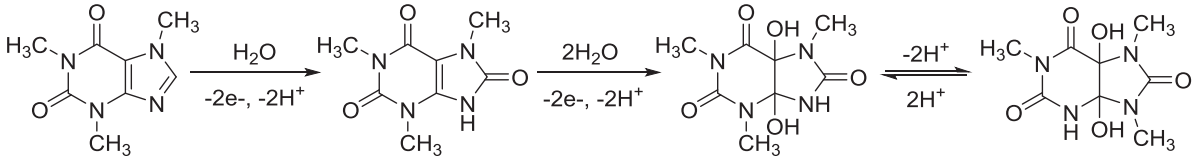
Standard deviations of these values are inconsiderable so repeatability of the method is adequate. Real height is slightly smaller compared to the models as a result of an initial distance between the nozzle and the bed which was lower than 0.3 mm to support adhesion of the scaffolds during the printing. Real porosity is somewhat different from designed 70% (for S1 and S2) or 50% (for S3 and S4). The extrusion width was 0.5 or 1.0 mm whereas values equal or lower than 0.5 mm are optimal for the used nozzle according to the producer. This could be improved by changing the standard nozzle. Nevertheless, accuracy of the method is sufficient. Microscopic images at 20x and 50x magnification are presented in Figure 1. White color is a result of homogeneous distribution of CAF in the polymer matrix. The structures are regular and fully reproduce the virtual models.

***Optimization of experimental conditions of DPV determination of CAF at GCE.—***To establish the most adequate experimental conditions for the DPV determination of CAF using the GCE, a univariate optimization study was carried out for parameters that may affect the voltammetric signal of the target species.

***Oxidation of CAF at GCE.—***The oxidation mechanism of caffeine was already elucidated by Hansen and Dryhurst in 1971.[39] The reaction proceeds by an overall four electrons and four protons process. The first reaction step is a two-electron and two-proton oxidation leading to the 1,3,7-trimethyluric acid. Next, $2e^-$ and $2H^+$ oxidation

**Table II. Parameters for the evaluation of printed scaffolds quality.**

| Model | Diameter [mm] | Height [mm] | Mass [g] | Porosity [%] |
|---|---|---|---|---|
| S1 | $15.08 \pm 0.07$ | $5.07 \pm 0.04$ | $0.659 \pm 0.008$ | $63.69 \pm 0.51$ |
| S2 | $15.12 \pm 0.10$ | $5.16 \pm 0.05$ | $0.853 \pm 0.014$ | $69.19 \pm 0.78$ |




**Scheme 1.** Electrochemical oxidation of caffeine (CAF).

follows with the formation of 4,5-diol analogues of methylated uric acid. Finally, rapid fragmentation occurs. The proposed mechanism for the electrochemical oxidation of CAF in aqueous solution occurs according to the mechanism adopted in literature and presented below in Scheme 1.

***Composition of supporting electrolyte.***—The choice of the supporting electrolyte is a very important stage in electro-analytical studies. Its composition and pH influence the properties of the analyzed solution and modify thermodynamics and kinetics of the charge transfer process. The effects of supporting electrolytes with different composition and pH values were investigated in order to choose the best medium for the DP voltammetric determination of CAF in analyzed sample solutions. The influence of the following supporting electrolytes at concentration of 0.05 mol $\cdot$ $L^{-1}$ on the peak potential (Ep) and peak current ($I_p$) of CAF was determined: sulfuric acid(VI), nitric acid(V), hydrochloric acid, acetic acid and acetate buffer. The measurements were carried out in solutions containing of 6 $\mu$mol $\cdot$ $L^{-1}$ CAF. According to this research the best shaped peak currents and lowest the background level of CAF were obtained in sulfuric acid(VI). Thus, further measurements were performed in a $H_2SO_4$ medium with the concentrations from 0.01 to 0.10 mol $\cdot$ $L^{-1}$ (Fig. 2A). The highest values of the peak currents with the best signal-to-background current ratio were obtained at the concentration of 0.05 mol $\cdot$ $L^{-1}$ which was adopted for the subsequent experiments.

***Optimization of the DPV parameters.***—DPV parameters were evaluated based on univariate experiments using 6 $\mu$mol $\cdot$ $L^{-1}$ of CAF in 0.05 mol $\cdot$ $L^{-1}$ $H_2SO_4$. The variables, including step potential ($E_s$), pulse amplitude (dE) and pulse width ($t_{imp}$) in the range of 1–6 mV, 20–70 mV and 10–50 ms respectively, were evaluated. Electrochemical cleaning of the electrode and surface regeneration is one of the main features of DPV. The cleaning step was optimized at −0.2 V for 3 s to restore the baseline and avoid electrode contamination from a previous measurement. Optimization of the time and potential of CAF pre-concentration on the GCE was not performed because CAF is not accumulated on its surface.

The influence of step potential on the peak current of CAF was investigated. As might be expected, higher $E_s$ values resulted in an increase in the peak current and the background level. Therefore, further studies were carried out using a step potential of 3 mV.

Following this study, an influence of the pulse amplitude (dE) upon the analytical signal was also investigated in positive and negative modes in the range between 70 and −70 mV. The magnitude of peak currents were similar for both positive and negative values. In Figure 2B, which shows the effect of the positive amplitude values on the height of the peaks among others, the linear relationship between these parameters could be seen ($R = 0.999$, $y = 0.048x + 1.37$, where y is the peak current (μA) and x is the pulse amplitude (mV)). Unfortunately, the background level was also found to increase with the increase of the pulse amplitude. Therefore dE of 50 mV connected with the best signal precision and signal-to-background ratio was used in all subsequent work.

The effect of pulse width ($t_{imp} = t_w$ (waiting time) + $t_s$ (current sampling time), $t_w = t_s$) from 10 to 50 ms was also examined (Fig. 2B). It was found that both oxidation peak values of CAF and background level decrease for longer pulse times. In view of these results, the pulse width of 20 ms was selected for further experiments.

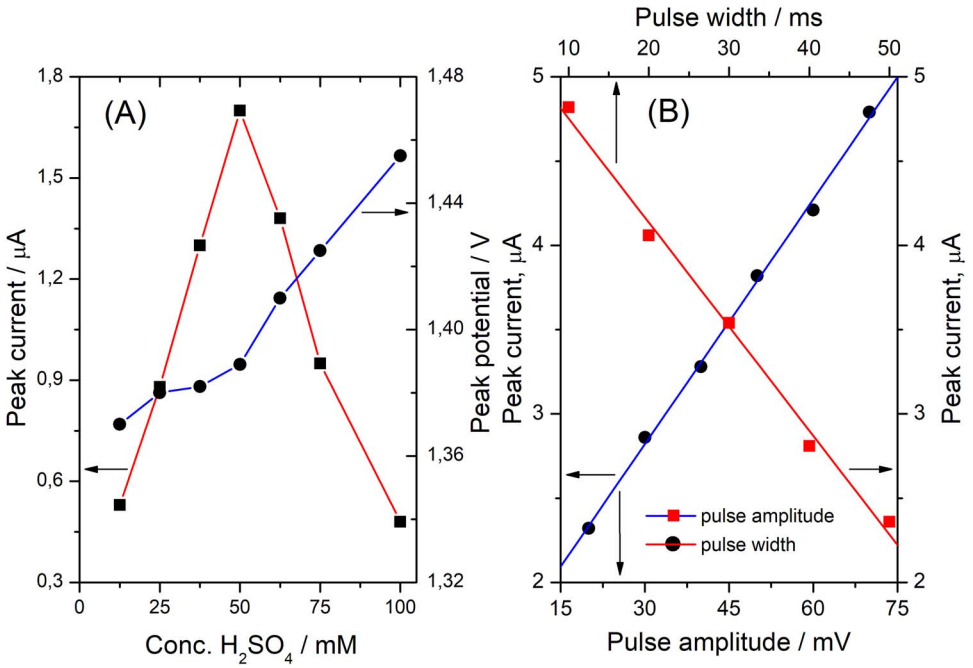



**Figure 2.** The influence of the following parameters on the peak current and potential for 6 $\mu$mol $\cdot$ $L^{-1}$ CAF: (A) concentration of sulfuric acid(VI) and (B) pulse amplitude and pulse width ($t_{imp}$ = waiting time = sampling current time). Activation potential and time: $E_{act} = -0.20$ V, $t_{act} = 3$ s. DP instrumental parameters: $t_{eq} = 3$ s, $E_s = 3$ mV, dE = 50 mV, $t_{imp} = 20$ ms.

**Table III. Optimal DPV technique parameters for the determination of CAF using GCE.**

| Parameter | Studied interval | Optimal value | Unit |
|---|---|---|---|
| Start potential ($E_{st}$) | 0.7–1.1 | 1.0 | V |
| Step potential ($E_s$) | 1–6 | 3 | mV |
| Pulse amplitude (dE) | 70–−70 | 50 | mV |
| Current sampling time ($t_s$) | 5–25 | 10 | ms |
| Waiting time ($t_w$) | 5–25 | 10 | ms |
| Pulse period ($t_{imp} = t_w + t_s$) | 10–50 | 20 | ms |
| $H_2SO_4$ concentration | 0.01–0.1 | 50 | mol · $L^{-1}$ |
| $t_{equilibration}$ ($t_{eq}$) | 1–15 | 3 | s |
| $E_{cleaning}$ ($E_{act}$) | −0.5–1.1 | −0.2 | V |
| $t_{cleanning}$ ($t_{act}$) | 0–30 | 3 | s |

Table III summarizes the optimized variables and the studied intervals.

***Stability of the GC working electrode.—***For practical application of the procedure long-term stability of the working electrode is very essential. It was evaluated by using the same GCE for 10 repetitive measurements in a supporting electrolyte solution containing 6 $\mu mol \cdot L^{-1}$ of CAF. The electrode used in this measurement was kept in DI water and subjected to the another 10 repetitive measurements after 24 h. No appreciable change in the peak potential and peak current for oxidation of CAF was noticed. The coefficients of variation in the peak current for both sets of experiments were calculated to be 1–3%, confirming that the GC working electrode is stable and can be used for the repeated measurements.

In order to further determine the operational stability of the working electrode, voltammetric measurements in a supporting electrolyte solution containing 6 $\mu mol \cdot L^{-1}$ of CAF were performed with GC electrode and the peak current for the oxidation of CAF was measured at regular intervals (2 h) over a period of 12 h. The magnitude of the peak current did not change appreciably during the whole set of experiments (12 h) demonstrating that GC electrode is very stable.

The long-term stability was tested by measuring the electrode response (peak current) for a period of one week. The response did not change appreciably and only 20% loss of activity was noticed after a storage time of one week in DI water at room temperature.

To check the reproducibility of the results, two different GC electrodes were used for the DPV measurement of a solution containing 6 $\mu mol \cdot L^{-1}$ of CAF. Both electrodes exhibited identical voltammograms for the oxidation of CAF, confirming that the results are highly reproducible. The standard deviation in the peak current and peak potential was calculated to be 0.2 μA and 3 mV, respectively.

***Application of the proposed DPV method in the analysis of CAF.—***The calibration graph for CAF at the GC electrode was linear from 1 and 30 $\mu mol \cdot L^{-1}$ and fulfill the equation $y = 0.33 \pm 0.01x$ - 0.09 ± 0.02, where $y$ is the peak current (μA) and $x$ is the CAF concentration ($\mu mol \cdot L^{-1}$). The correlation coefficient $R$ was 0.9998. Received DP voltammograms and linear range of the calibration graph are presented in Figures 3A and 3B, respectively. For the repeatability study, three levels of CAF concentration, 6, 14 and 22 $\mu mol \cdot L^{-1}$, were evaluated (n = 5). Relative standard deviations (RSD) were 3.1%, 2.9% and 3.3%, respectively. These values show the efficiency of the electrochemical cleaning step to avoid contamination of the GCE from a previous measurement. The values of limit of detection (LOD) and limit of quantification (LOQ) were 0.24 and 0.73 $\mu mol \cdot L^{-1}$, respectively (LOD = 3sB/S and LOQ = 10sB/S, where sB is the standard deviation of baseline noise for n = 5 and S is the slope of analytical curve).

***Tolerance to interfering species.—***Selectivity of the proposed DPV procedure was estimated using the additions of potentially interfering ions (Cr, Cu, Co, Fe, Ni, Pb, Sn and Zn) to a solution containing 5 $\mu mol \cdot L^{-1}$ CAF. Except for Cu and Fe other ions have negligible interference. Addition of Cu(II) and Fe(III) at a concentration of 10 $mmol \cdot L^{-1}$ induced a decrease of CAF signal to 91.2% and 82.6% of its original value, respectively. It should be noted that the analytical signal of CAF was still well-formed and easy to measure.

What is more, the influence of surfactants on the peak current for 5 $\mu mol \cdot L^{-1}$ CAF was investigated by the addition of standard surface active substances (SAS): Triton X-100 (neutral SAS), CTAB (cationic SAS) and SDS (anionic SAS) to the studied solution in the range from 0 to 2.0 $mg \cdot L^{-1}$. It was observed that the additions of 2.0 $mg \cdot L^{-1}$

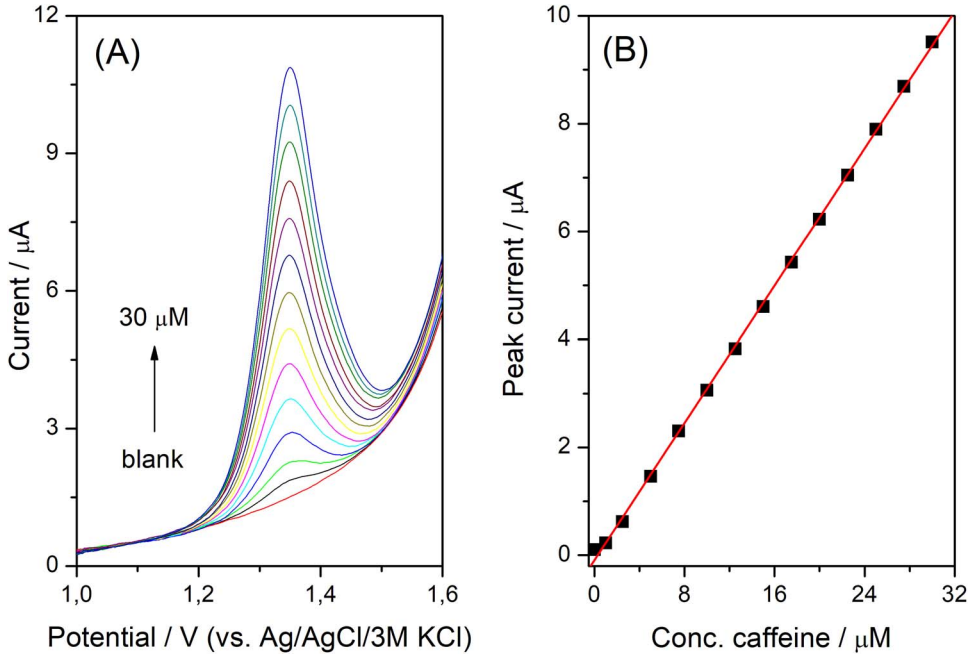



**Figure 3.** (A) DP voltammograms obtained at the GCE in a 0.05 $mol \cdot L^{-1}$ $H_2SO_4$ solution and increasing concentrations of CAF (from bottom to top): blank, 1, 2.5, 5, 7.5, 10, 12.5, 15, 17.5, 20, 22.5, 25, 27.5, 30 $\mu mol \cdot L^{-1}$. (B) Corresponding calibration plot. Other conditions as in Fig. 2.

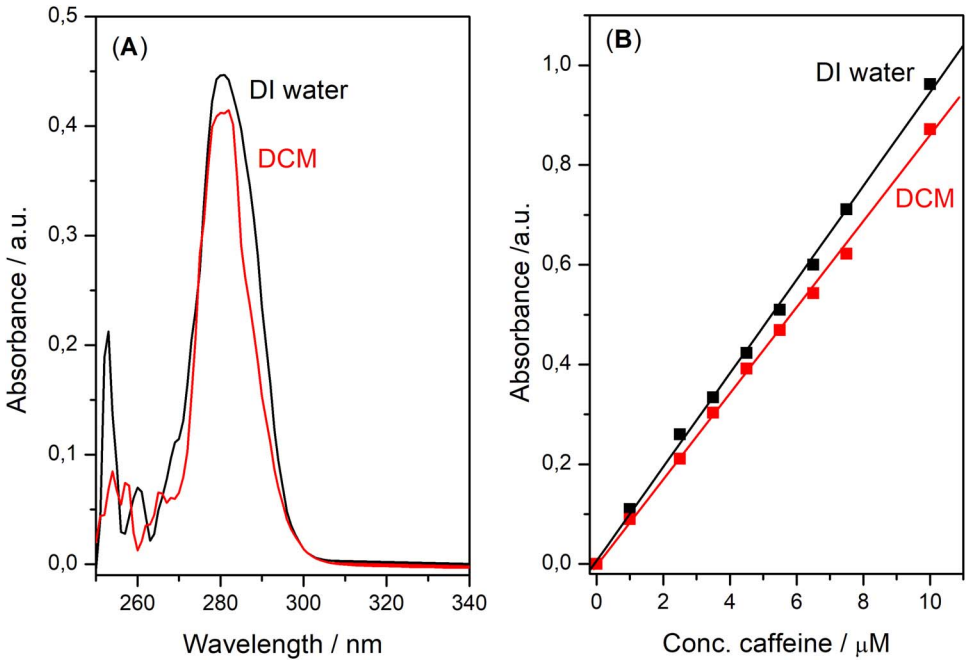



**Figure 4.** (A) UV-Vis spectra of 5 $\mu mol \cdot L^{-1}$ CAF in DI water and dichloromethane (DCM). (B) Corresponding calibration plots.

Triton X-100 and SDS increased the height of the CAF signal by 6% and 1% of its original value respectively, while the addition of 2.0 $mg \cdot L^{-1}$ CTAB decreased the height of the CAF signal by 29% of its original value. In all cases, the caffeine signal was well-defined. The above data indicate that use of the proposed voltammetric method and GC electrode allowed for the determination of CAF without the need of sample pretreatment step in extracts that could be contaminated by the trace amounts of organic compounds from the preparation process of the samples (greases etc.).

In CAF diluted solutions ($<10^{-5}$ $mol \cdot L^{-1}$), its slow decay due to Vis radiation was observed. As a result, after 2–3 hours of exposition, the current of CAF oxidation decreases of 5–10%, depending on Vis radiation intensity. Nevertheless, CAF is chemically stable in various conditions and environments.[46,47] Certain enzymes and bacteria can decompose CAF, leading to formation of five metabolites.[48,49]

***UV-Vis spectra of CAF in water and dichloromethane.***—UV-Vis absorption spectra of 5 $\mu mol \cdot L^{-1}$ CAF in DI water and dichloromethane (DCM) were measured between 240 and 340 nm at the room temperature (Fig. 4A). For both solvents, the maximum absorbance was recorded at the wavelength of $\lambda_{max} = 278$ nm.

The peak absorbance of CAF was found to be A = 0.447 and A = 0.413 in 5 $\mu mol \cdot L^{-1}$ solutions in DI water and DCM, respectively. The molar absorption coefficient at a given wavelength was calculated using Beer-Lambert's equation. The molar decadic absorption coefficients of CAF were also calculated and value of $\varepsilon_{max} = 17880$ $dm^3 \cdot mol^{-1} \cdot cm^{-1}$ and $\varepsilon_{max} = 16520$ $dm^3 \cdot mol^{-1} \cdot cm^{-1}$ were obtained for DI water and DCM, respectively.

The maximum peak absorbance of CAF observed in this experiment was quite similar to those reported by literature.[44]

***Validation of UV-Vis method.***—UV-Vis method recommended by European pharmacopoeia (9th ed., Strasbourg, Council of Europe, 2016) was evaluated for 5 $\mu mol \cdot L^{-1}$ CAF solution under the selected experimental conditions: $\lambda_{max} = 278$ nm, 2 nm slit width and 1 cm quartz cuvette by checking calibration range, accuracy and precision. The calibration plots were linear in the range: 1 – 10 $\mu mol \cdot L^{-1}$, $y = 0.094 \pm 0.001x - 0.006 \pm 0.001$ ($R = 0.999$) and $y = 0.085 \pm 0.001x - 0.002 \pm 0.001$ ($R = 0.999$) for DI water and dichloromethane, respectively. The standard addition principle (the model samples spiked with various known amounts of CAF) was used to evaluate the precision (characterized by the relative standard deviation percentage, RSD%) and accuracy (characterized by the recovery percentage, R%) of the recommended method. The results are given in Table IV.

***Spectrophotometric UV-Vis and voltammetric DPV techniques profiles of CAF release from scaffolds.***—CAF release (extraction) profiles of 3D PLA/CAF scaffolds were designated for three different diluents: DI water, 0.1 $mol \cdot L^{-1}$ PBS (pH 7.4) and 0.1 $mol \cdot L^{-1}$ HCl, independently for UV-Vis and DPV techniques. Individual samples of CAF extracts were obtained after 1, 2, 4, 24, 48, 72 and 96 hours of conditioning scaffolds in diluents in a manner and conditions as described in Solid-liquid extraction of CAF section. Three scaffolds were placed in each of three diluents, thus obtaining 9 independent analytical samples. Samples of the extracts were not stored but analyzed immediately without the pretreatment. They were only diluted with DI water or supporting electrolyte (for DPV) to the concentration that provides the best precision and accuracy of the analysis. Concentration of CAF in the extract using UV-Vis technique was determined by the interpolation method based on the calibration curve of Figure 4B, while for DPV technique by 3 standard additions method. In case of DPV, the use of interpolation method is not possible, as the sensitivity of the working electrode can change from one solution to another, thus voltammetric methods cannot be standardized as for example UV-Vis spectrometry. Each extract was analyzed three times and the results were averaged. Exemplary CAF oxidation voltammograms without baseline correction after 24 h of scaffolds conditioning in diluents:

**Table IV. Precision and accuracy data for the UV-Vis determination of CAF in DI water and dichloromethane (n = 3).**

| Solvent | Added [$\mu mol \cdot L^{-1}$] | Found [$\mu mol \cdot L^{-1}$] | RSD [%] | Recovery [%] |
|---|---|---|---|---|
| DI water | 3.0 | 2.9 | 3.2 | 96.7 |
| | 5.0 | 5.3 | 2.9 | 106.0 |
| | 7.0 | 6.9 | 3.5 | 98.6 |
| Dichloromethane | 3.0 | 2.9 | 3.6 | 96.7 |
| | 5.0 | 4.8 | 4.1 | 96.0 |
| | 7.0 | 6.8 | 3.7 | 97.1 |

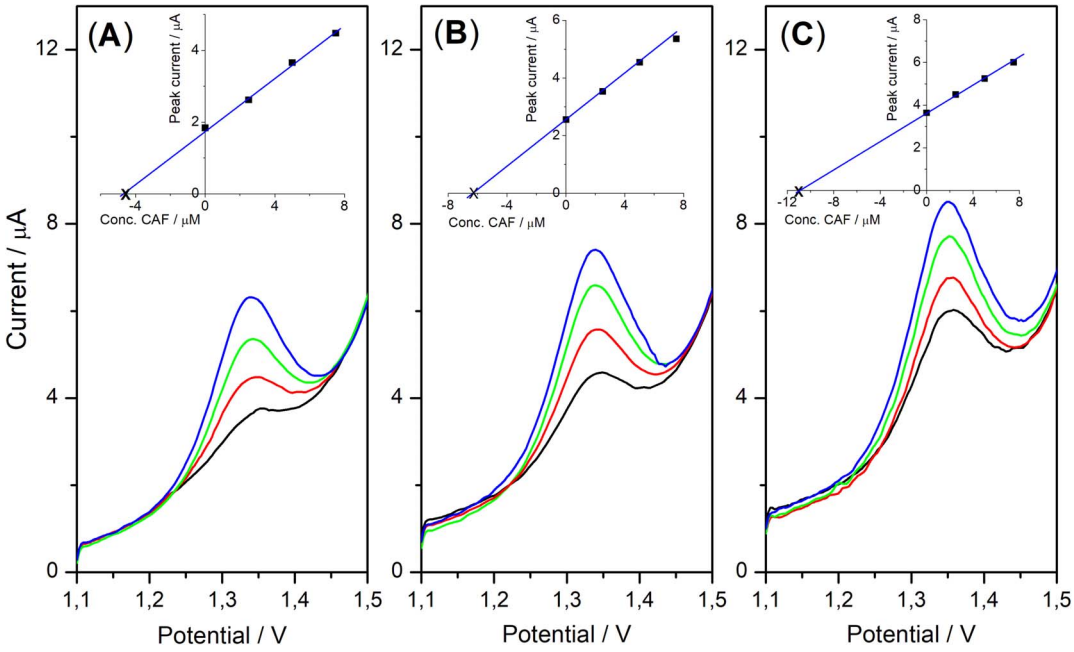



**Figure 5.** Quantification of CAF in the extract samples after 24 hours of scaffolds conditioning in solutions: (A) 0.1 mol · $L^{-1}$ HCl, (B) 0.1 mol · $L^{-1}$ PBS (pH 7.5) and (C) DI water. Inset: corresponding standard additions plots. Curves (from bottom to top): extract sample with addition of the supporting electrolyte; 2.5; 5.0 and 7.5 µmol · $L^{-1}$ CAF (3 repetitions). All other conditions as in Fig. 2.

0.1 mol · $L^{-1}$ HCl, 0.1 mol · $L^{-1}$ PBS (pH 7.4) and DI water, are presented in Figure 5. The obtained correlations for standard additions plots were excellent.

Figure 6 illustrates long-term CAF release profiles in the diluents obtained for the same sample using UV-Vis spectrophotometry (Fig. 6A) and DPV technique (Fig. 6B). For example, CAF concentration after 2 hours of extraction was calculated by summing the determined amounts of CAF (in mg) extracted after the first and second hour of the scaffolds conditioning and concentration after 4 hours of extraction was calculated by summing the amounts obtained after first, second and fourth hour of the conditioning.

Based on Figure 6, it is clear that in a time interval from 1 to 24 hours of CAF release from the scaffolds, both tested methods, i.e. UV-Vis and DPV, show similar sensitivity and both are equally useful. But for a longer release times, i.e. in the range from 24 to 96 h, significantly better selectivity and higher sensitivity of the CAF assay were obtained for the DPV technique. It can be concluded by the analysis of the profiles for UV-Vis technique (Fig. 6A) that the process of CAF release from the scaffolds reaches a negligibly low efficiency after 24 h for 0.1 mol · $L^{-1}$ HCl, while after 48 and 72 h for 0.1 mol · $L^{-1}$ PBS and DI water respectively. The profiles obtained for DPV technique (Fig. 6B) indicate that from 4 to 48 h the CAF release

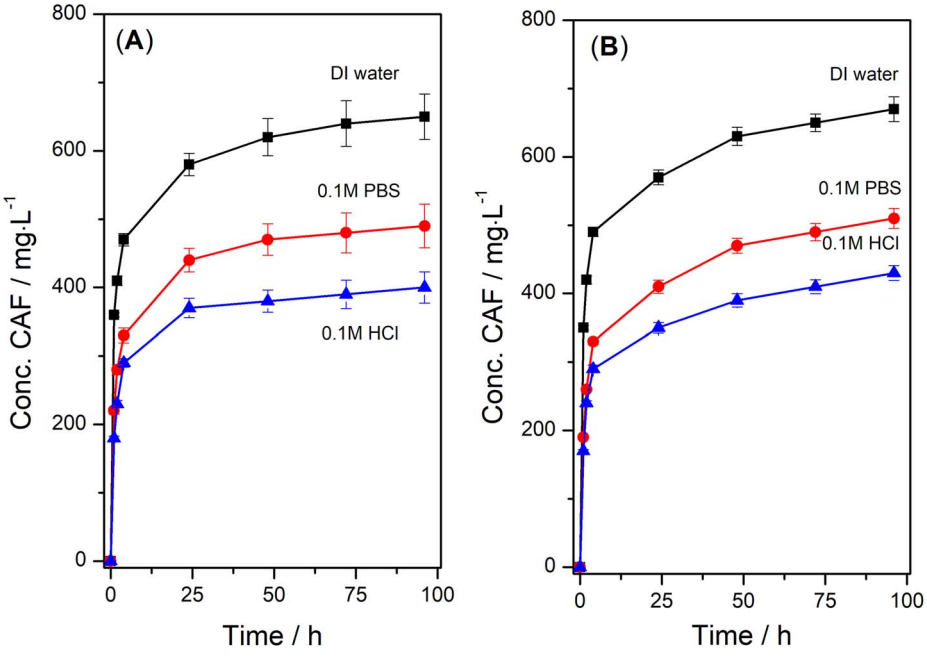



**Figure 6.** CAF release (extraction) profiles the diluents: DI water, 0.1 mol · $L^{-1}$ PBS (pH 7.4) and 0.1 mol · $L^{-1}$ HCl obtained by UV-Vis (A) and DPV (B). Scaffolds conditioning time in diluents was 96 h, volume of the diluent was 10 mL and approximate mass of a single scaffold was 0.65 g. Diluents were changed after: 1, 2, 4, 24, 48, 72 and 96 hours of conditioning.

**Table V. Amounts of extracted CAF per hour, obtained for UV-Vis and DPV techniques, (n = 3).**

| | | Amount of extracted caffeine per hour [mg · L$^{-1}$], RSD [%] Conditioning time [h] | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Methods | Solutions | 1 | 2 | 4 | 24 | 48 | 72 | 96 |
| UV-Vis | DI water | 362.0 (0.8) | 52.5 (1.7) | 31.0 (1.9) | 5.6 (2.8) | 1.7 (4.4) | 0.5 (5.2) | 0.4 (5.1) |
| | PBS | 221.0 (2.3) | 63.5 (2.6) | 25.5 (3.3) | 5.3 (3.9) | 1.3 (4.9) | 0.5 (6.0) | 0.4 (6.5) |
| | HCl | 183.5 (1.5) | 49.5 (1.5) | 32.0 (2.1) | 4.4 (3.8) | 0.9 (4.2) | 0.4 (5.3) | 0.3 (5.7) |
| DPV | DI water | 351.5 (0.9) | 72.0 (1.4) | 35.5 (1.3) | 4.1 (1.9) | 2.8 (2.1) | 0.9 (2.0) | 0.7 (2.7) |
| | PBS | 193.0 (1.0) | 71.5 (1.3) | 34.5 (1.2) | 4.3 (2.4) | 2.4 (2.3) | 0.8 (2.6) | 0.8 (2.9) |
| | HCl | 172.5 (1.3) | 78.5 (1.5) | 22.0 (1.9) | 3.6 (2.2) | 1.7 (2.5) | 0.6 (2.5) | 0.6 (2.5) |

The RSD values in % are given in the brackets.

process proceeds with an average yield of about 3.0 mg · h$^{-1}$ and later with a yield of about 0.8 mg · h$^{-1}$. Table V summarizes the results of the determination of CAF in the individual extract.

Considering various diluents but always with the same volume (10 mL), identical extraction conditions and similar scaffolds (about 0.65 g) it was found that after 96 hours of extraction with 3D PLA/CAF scaffolds 1.03%, 0.75% and 0.65% of CAF were released into DI water, 0.1 mol · L$^{-1}$ PBS (pH 7.4) and 0.1 mol · L$^{-1}$ HCl respectively.

## Conclusions

In this work, differential pulse voltammetry (DPV) technique was used for the first time in bioengineering, specifically in the study of degree and efficiency of caffeine release from bone tissue implants produced by 3D printing technique. The methodology of the preparation of PLA/CAF filament (mass ratio 9:1) and 3D scaffolds printing process were developed and described in detail. Even for 10% of CAF share in PLA filaments, excellent accuracy in the reproduction of the printed structures was achieved. The release of caffeine from PLA/CAF scaffolds was tested in DI water, 0.1 mol · L$^{-1}$ PBS (pH 7.4) and 0.1 mol · L$^{-1}$ HCl. It was found that DPV technique demonstrates much better selectivity and sensitivity of CAF determination in relation to UV-Vis spectrophotometry recommended and commonly used in bioengineering. In addition, the linearity range for DPV technique is nearly four times wider, which significantly simplifies and speeds up the procedure of preparation (dilution) of extract samples for the analysis. The versatility of DPV technique and the ability to effectively concentrate various drugs on the surface of the working electrode makes it willingly used in their analysis. Thus, we provide perspectives on the use of known and validated voltammetric drug analysis procedures in the field of bioengineering of biologically active implants.

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